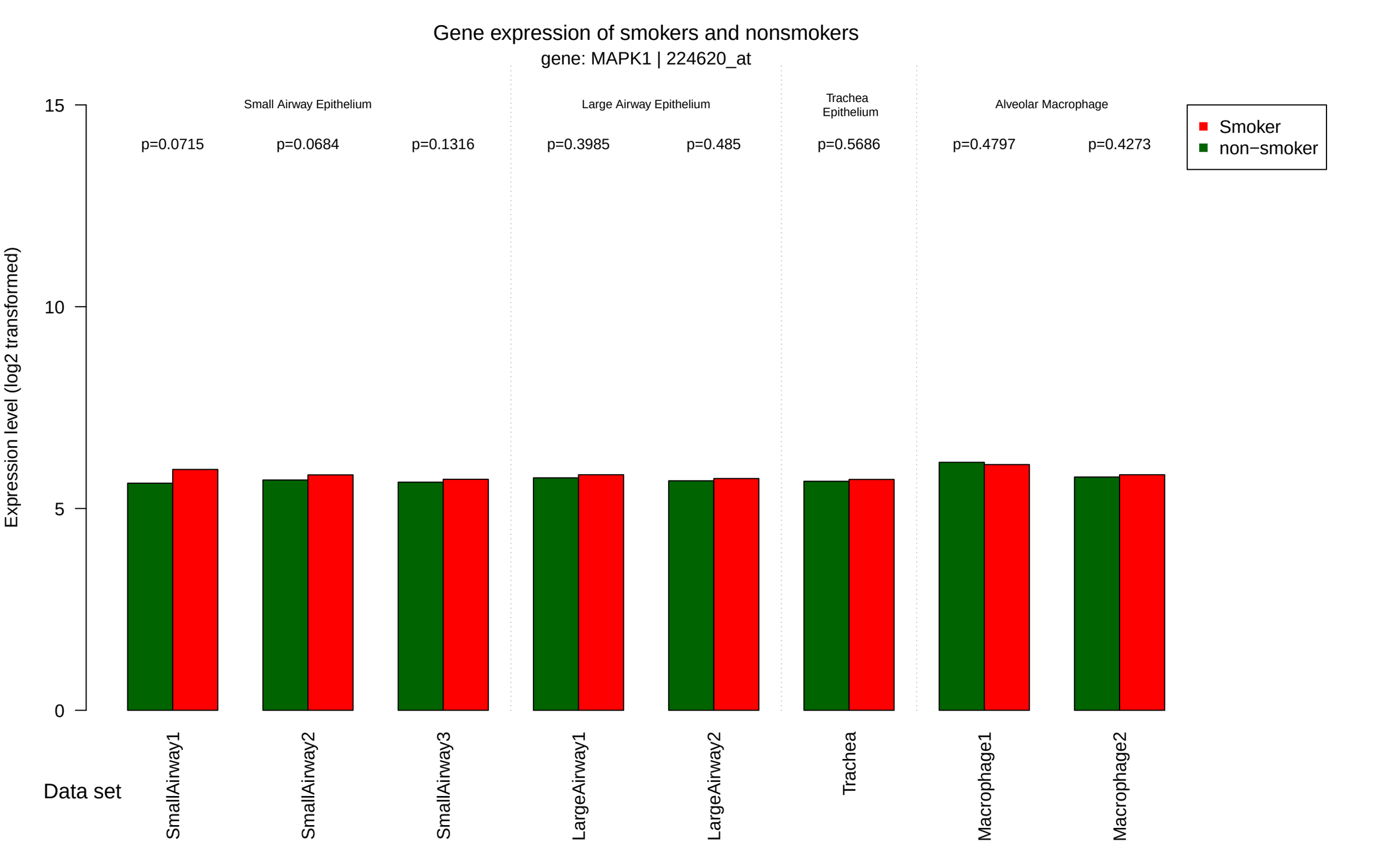
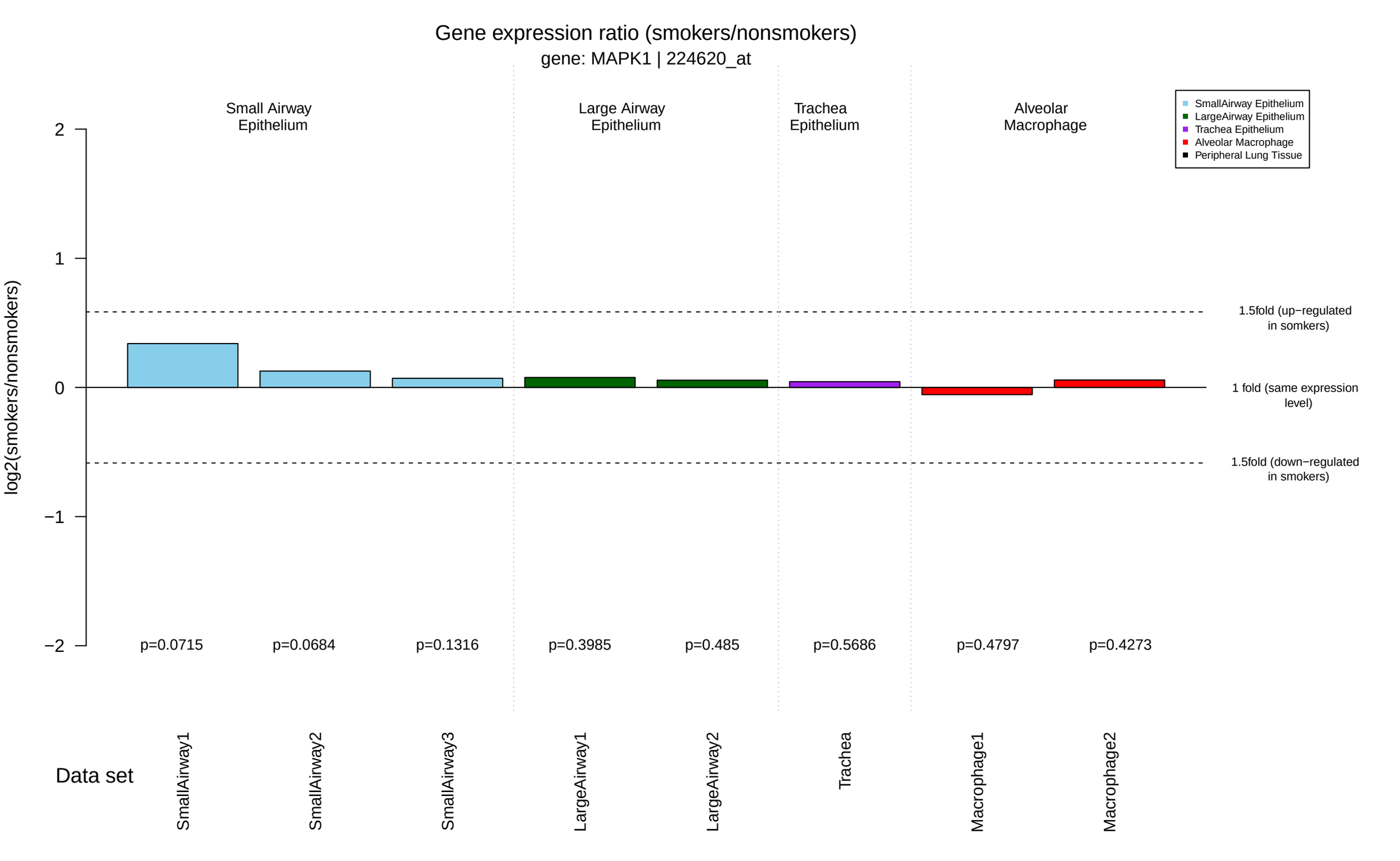
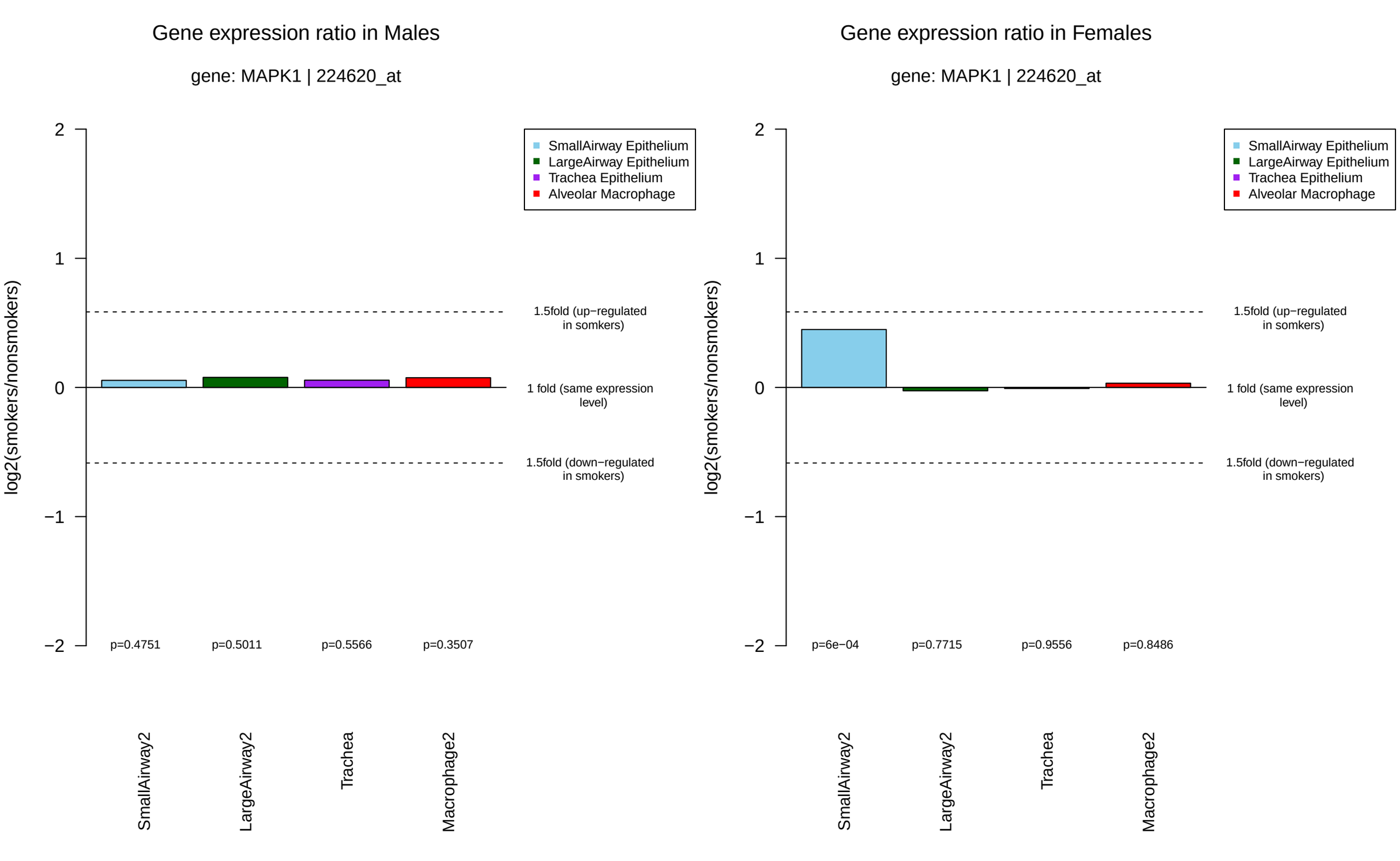
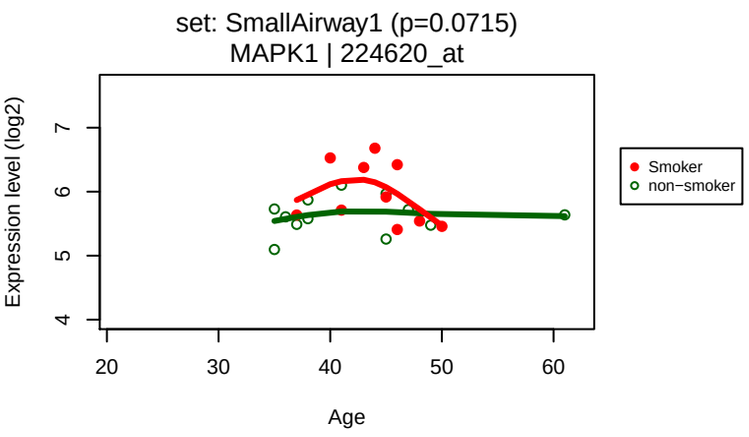
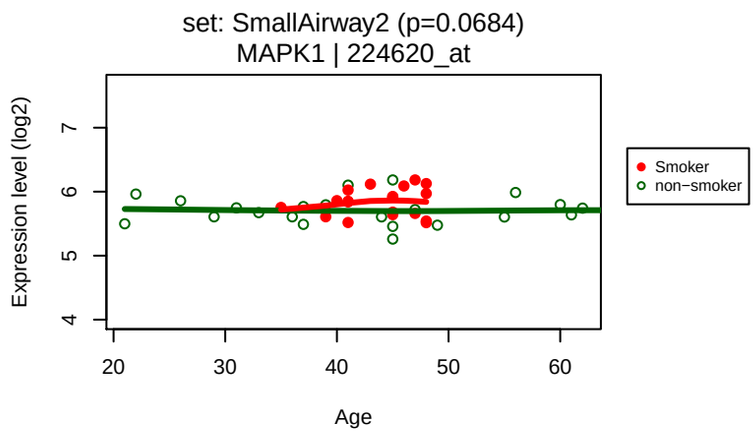
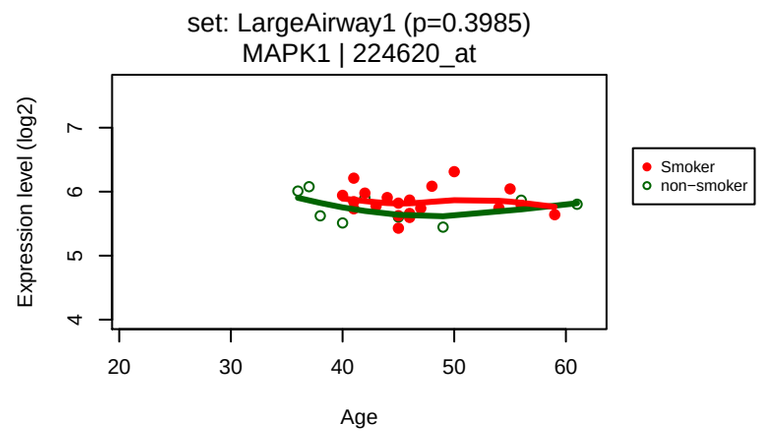

Gene expression of smokers and nonsmokers
gene: MAPK1 | 224620_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0715
p=0.0684
p=0.1316
p=0.3985
p=0.485
p=0.5686
p=0.4797
p=0.4273
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

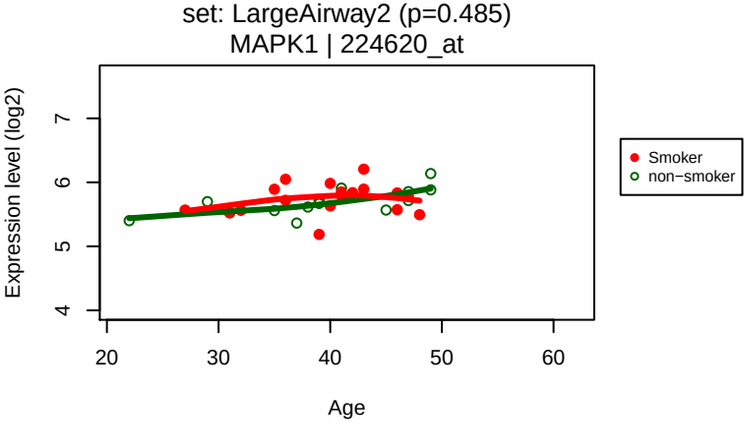

Gene expression ratio (smokers/nonsmokers)
gene: MAPK1 | 224620_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0715
p=0.0684
p=0.1316
p=0.3985
p=0.485
p=0.5686
p=0.4797
p=0.4273
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

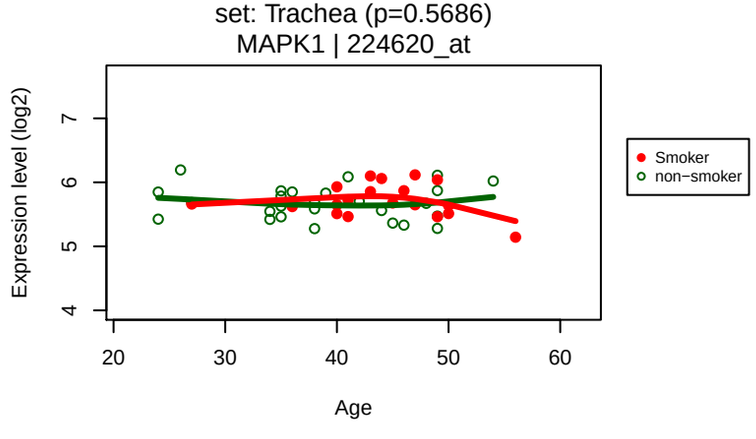

Gene expression ratio in Males
gene: MAPK1 | 224620_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4751
p=0.5011
p=0.5566
p=0.3507
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAPK1 | 224620_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=6e−04
p=0.7715
p=0.9556
p=0.8486
SmallAirway2
LargeAirway2
Trachea
Macrophage2

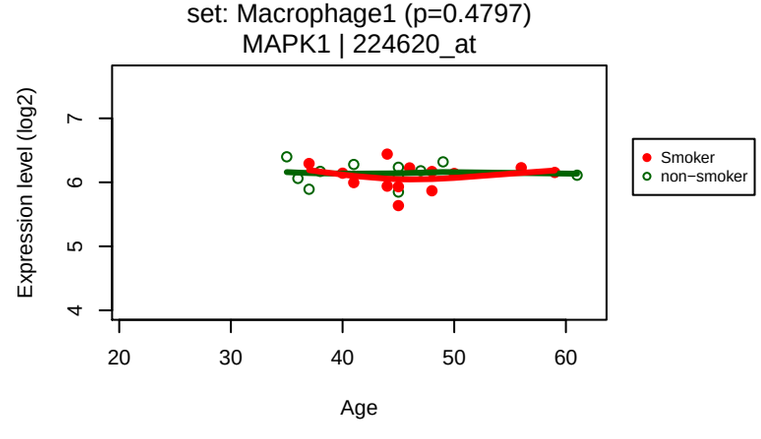

set: SmallAirway1 (p=0.0715)
MAPK1 | 224620_at
Expression level (log2)
Age
Smoker
non−smoker

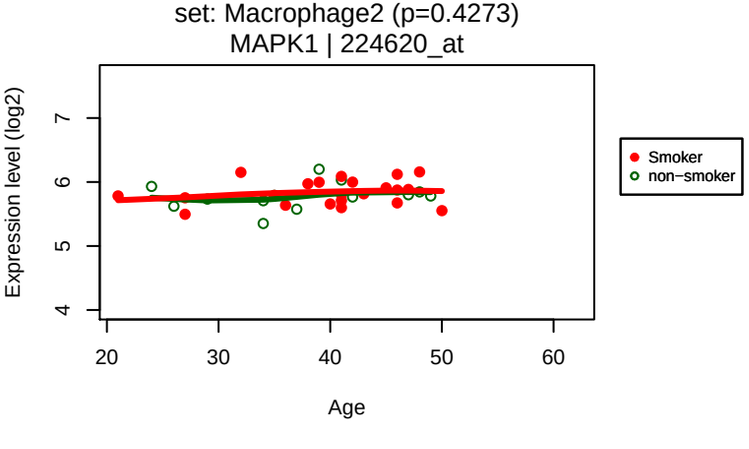

set: SmallAirway2 (p=0.0684)
MAPK1 | 224620_at
Expression level (log2)
Age
Smoker
non−smoker

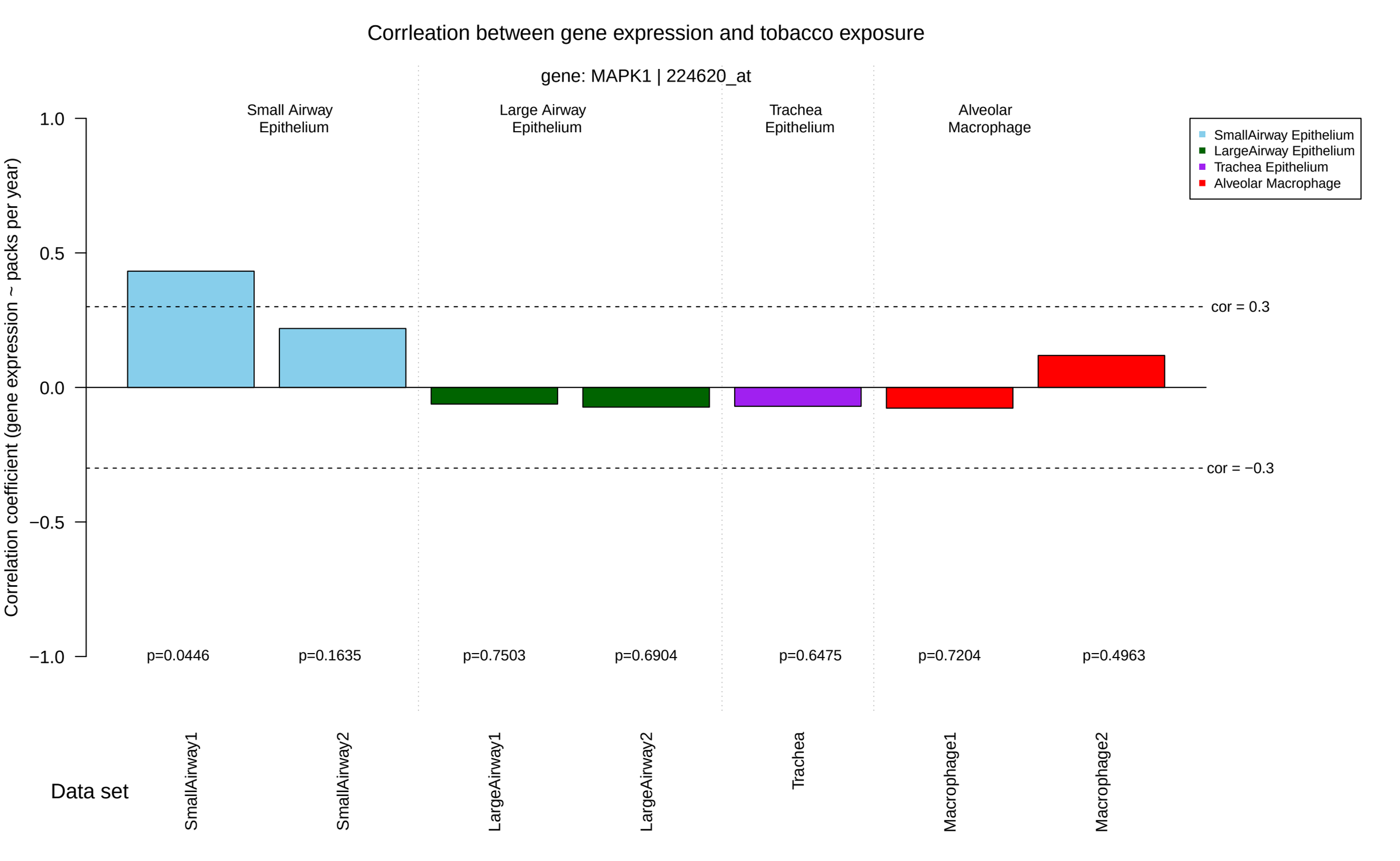

set: LargeAirway1 (p=0.3985)
MAPK1 | 224620_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.485)
MAPK1 | 224620_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5686)
MAPK1 | 224620_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4797)
MAPK1 | 224620_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4273)
MAPK1 | 224620_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAPK1 | 224620_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0446
p=0.1635
p=0.7503
p=0.6904
p=0.6475
p=0.7204
p=0.4963
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2